Solve Problems with Unit Rates

Write cost over quantity, find the unit rate, and choose the better seller.

Name _____ **Date** _____

Tonight's spark

A team needs 12 notebooks. Shop A sells 4 notebooks for $10.00. Shop B sells 6 notebooks for $13.80. Which shop costs less for 12 notebooks?

Make a guess first: is Shop A or Shop B cheaper?

Notebooks (Shop B)	Cost (Shop B)
6	$13.80
1	?
12	?

÷6

×12

Our class model: cost on top, quantity on the bottom

Try it

Work it out on paper. Write one answer on each line.

1. **A sub shop sells 2 subs for $13. What is the cost of 1 sub?**

   A. $11 **B.** $26 **C.** $6.50 **D.** $15

   My answer _____

2. **A team needs 12 water bottles at $1.25 each. How much do 12 bottles cost?**

   A. $15 **B.** $13.25 **C.** $9.60 **D.** $10.75

   My answer _____

3 **Seller 1 charges $3.50 per hat. Seller 2 charges $18 for 6 hats. Which seller is the better deal?**

A. Seller 1, because $3.50 is less than $18 **B.** Seller 2, $3 per hat

C. Seller 1, $3.50 per hat **D.** They are the same

My answer _____

4 **The better seller charges $0.35 per pencil. What is the total for 20 pencils?**

A. $0.02 **B.** $20.35 **C.** $7.00 **D.** $70.00

My answer _____

5 **8 melons cost $20. Divide both parts by 8. What is the cost of 1 melon?**

A. $12 **B.** $160 **C.** $0.40 **D.** $2.50

My answer _____

Talk about it

Which seller has the lower unit rate, and how did you use it to price the whole order?

Home mission

Find two shelf tags for the same item in different sizes. Write each as cost over amount, like $13.80 over 6. Divide the top and the bottom by the bottom number. Which is cheaper for 1?

If we get stuck, ask

- Cost on TOP, amount on the BOTTOM. Now divide both parts by the bottom number.
- The unit rate is a fraction over 1 (like 2.50/1). What does each number tell us?
- Why can the bigger total still be the cheaper deal per item?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/3-8-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it